INTRODUCTION TO NUMBER SYSTEMS

PART 2

DECIMAL TO BINARY CONVERSION

RECAP

1. The decimal number system uses base 10 0,1,2,3,4,5,6,7,8,9

2. The binary number system uses base 2 consists of only two digits: 0 and 1.

Conversion means representing the same value in a different number system.

Example:

Decimal	Binary
5	101_2

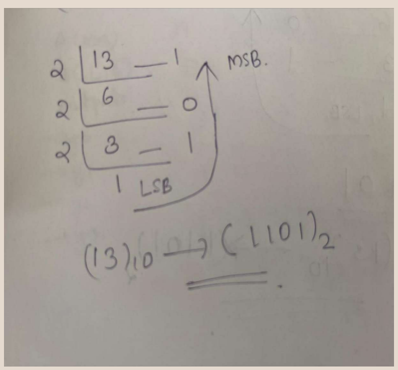
Both numbers represent the same quantity, but they are written using different number systems.

Method 1: Repeated Division by 2

Steps:

1. Divide the decimal number by 2.
2. Note the remainder.
3. Continue dividing the quotient by 2.
4. Stop when the quotient becomes 0.
5. Read the remainders from bottom to top.

Example: $13_{10} \rightarrow (\quad)_2$



$13_{10} \rightarrow (1101)_2$

Method 2: positional weight method

Example: 13_{10}

2^0	1
2^1	2
2^2	4
2^3	8
2^4	16

b0
b1
b2
b3
b4

13 <16

8<13

8+5 = 13

4 + 1=5

8+4+1 = 13

8 4 1

2^3 2^2 2^0

1 1 0 1

$13_{10} \rightarrow (1101)_2$

25.625_{10}:

- Integer Part = 25
- Fractional Part = 0.625

$25_{10} = 11001_2$

$0.625_{10} = 0.101_2$

$25.625_{10} = 11001.101_2$

Practice questions

$18_{10} = \ ?_2$
$31_{10} = \ ?_2$
$45_{10} = \ ?_2$
$67_{10} = \ ?_2$
$72_{10} = \ ?_2$
$100_{10} = \ ?_2$
$129_{10} = \ ?_2$
$255_{10} = \ ?_2$

$25.625_{10} = ?_2$
$50.125_{10} = ?_2$

3 bit binary

Decimal Number	Binary Representation
0_{10}	0_2
1_{10}	1_2
2_{10}	10_2
3_{10}	11_2
4_{10}	100_2
5_{10}	101_2
6_{10}	110_2
7_{10}	111_2
8_{10}	1000_2
9_{10}	1001_2

4 bit binary

0_{10}	0000_2
1_{10}	0001_2
2_{10}	0010_2
3_{10}	0011_2
4_{10}	0100_2
5_{10}	0101_2
6_{10}	0110_2
7_{10}	0111_2
8_{10}	1000_2
9_{10}	1001_2
10_{10}	1010_2
11_{10}	1011_2
12_{10}	1100_2
13_{10}	1101_2
14_{10}	1110_2
15_{10}	1111_2

Decimal	Hexadecimal
10_{10}	A_{16}
11_{10}	B_{16}
12_{10}	C_{16}
13_{10}	D_{16}
14_{10}	E_{16}
15_{10}	F_{16}

thank you